PROVISIONAL PATENT APPLICATION

Paired-Endpoint Direct-Adjacency Apparatus, Challenge-Bound Completion Protocol, Four-Ledger Custody Verification, and Methods of Qualifying and Operating the Same

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Filing type: U.S. Provisional Application under 35 U.S.C. §111(b)

This specification is complete and self-contained. Companion applications are nonessential background only. Every example under PROPHETIC EXAMPLES is prospective; no nonlocal completion, direct adjacency, custody transfer, payload transport, or distance-independent tariff is represented as achieved.

Portfolio prophetic-example statement: All examples identified as prophetic are prospective; no prophetic example is represented as an experiment performed or a result achieved.

CROSS-REFERENCE TO RELATED APPLICATIONS

This application belongs to a coordinated series concerning controlled angular-momentum coherence, completion-address engineering, and boundary control. Companion applications concern signaling and completion-scheduled computation. This application is distinct in requiring two physical endpoint assemblies, a fresh challenge bound to a terminal address record, paired endpoint receipts, ordinary-channel isolation, and a four-ledger custody reconciliation before any transport classification.

FIELD

Paired endpoint apparatus; address-selective transactions; authenticated physical challenge-response; isolated terminal event detection; custody-preserving transfer; boundary energy and momentum accounting; and null-valid methods for distinguishing direct adjacency from signaling, coincidence, delayed release, duplication, substitution, and destructive reconstruction.

BACKGROUND

1. Optical observation can identify a destination without providing physical access to it. Conventional transport traverses an ordinary route and incurs delay, energy, exposure, and custody along that route.
2. Communications systems, synchronized clocks, entangled-state experiments, remote actuation, teleportation protocols, and distributed databases can correlate endpoints or reproduce a state description. They do not by themselves establish transport of one continuously custodied physical organization.
3. Adding an edge between two nodes of a graph shortens their graph distance. That theorem is exact mathematics, but it does not tell an engineer how a material boundary creates the edge.

4. The applicant supplies a staged apparatus whose smallest experiment is not a vehicle or human portal. Two isolated endpoint assemblies exchange a fresh terminal challenge through a declared low-bandwidth authentication channel, then attempt a separate completion available only under the challenge-bound physical state. All ordinary electromagnetic, acoustic, mechanical, thermal, optical, software, timing, shared-environment, and hidden-carrier paths are measured or bounded.
5. A result is classified through four synchronized ledgers: source, terminal, support, and custody. A source anomaly without a terminal receipt is local. A terminal event without custody is communication or coincidence. A custody record without a closed support account is incomplete. Payload transport begins only after all four reconcile.

SUMMARY

A paired-endpoint apparatus (100; FIG. 1) includes source and terminal boundary assemblies, authenticated address registries, a challenge generator, phase and timing systems, completion actuators, isolation witnesses, custody stores, support-ledger sensors, and a trial controller. The terminal generates a fresh unpredictable challenge; the source binds its attempted state to that challenge and the selected terminal address; the endpoint controllers execute a randomized active or control trial; and an adjudicator classifies the synchronized record.

The first buildable rung is an address-selective terminal transaction experiment using extant resonators, clocks, challenges, and isolated detectors. It can return a useful null or identify hidden coupling. Higher rungs progress through paired receipt, direct adjacency, custody transfer of a retained state, and payload transport.

BRIEF DESCRIPTION OF THE DRAWINGS

- FIG. 1 — Source and terminal endpoint assemblies, challenge channel, attempted completion channel, witnesses, and adjudicator.
- FIG. 2 — Challenge-bound address record and trial transcript.
- FIG. 3 — Four-ledger source, terminal, support, and custody reconciliation.
- FIG. 4 — Randomized trial state machine and blinded label release.
- FIG. 5 — Ordinary-channel isolation enclosure and independent witness instruments.
- FIG. 6 — Countermodel matrix for hidden channel, duplicate, delayed release, substitution, destruction/reassembly, and clock artifact.
- FIG. 7 — Empirical adjacency map, tariff surface, validity envelope, and safe release.
- FIG. 8 — Progression from receipt to adjacency, custody, instrument package, and payload.
- FIG. 9 — Paired aperture modules and capacity-limited admission controller.
- FIG. 10 — Qualification-to-deployment, provenance, validity states, and independent confirmation.

DEFINITIONS

1. **Endpoint assembly** — a physical source or terminal boundary apparatus having an authenticated address record, local clock, completion actuator or detector, support sensing, and trial logger.
2. **Address record** — authenticated operational conditions identifying an endpoint, including hardware identity, state preparation, timing window, challenge key, admissible channel, and validity state.
3. **Fresh terminal challenge** — an unpredictable value generated at the terminal after the trial schedule is committed and bound to one attempted completion window.
4. **Challenge-bound preparation** — a measured source and terminal physical state whose phase, modulation, code, mode, polarization, sequence, or other completion condition depends on the fresh challenge.
5. **Declared authentication channel** — an ordinary bounded channel permitted to carry address and challenge information but not the claimed terminal energy, matter, or custody event.
6. **Attempted completion channel** — the separately instrumented physical operation under test.
7. **Paired receipt** — time-bound source and terminal records satisfying a frozen relation under the fresh challenge and absent under matched wrong-address, wrong-phase, dummy-load, and isolation controls.
8. **Direct adjacency** — a qualified paired receipt whose delay or tariff does not require traversal of the ordinary displayed route and which survives the disclosed countermodel gates. It does not by itself establish payload custody.
9. **Source ledger** — source preparation, state change, energy, momentum, angular momentum, timing, residual custody, and release record.
10. **Terminal ledger** — terminal challenge, readiness, event, energy, momentum, angular momentum, timing, and acquired custody record.
11. **Support ledger** — all permitted and witnessed ordinary channels, stored energy, heat, control traffic, environmental changes, and boundary costs.
12. **Custody ledger** — unique identity and retained-invariant record establishing whether one organization, rather than information or a duplicate, changed endpoint custody.
13. **Four-ledger reconciliation** — joint acceptance of the source, terminal, support, and custody ledgers under frozen tolerances.
14. **Custody transfer** — source custody is extinguished or lawfully handed off as terminal custody of the same retained organization is established, with no simultaneous duplicate and with declared invariants preserved.
15. **Tariff surface** — measured energy, time, preparation, phase, aperture, payload complexity, distance, mismatch, and support cost of an operation, including uncertainty and null regions.
16. **Countermodel** — a hidden ordinary channel, pre-positioned duplicate, coordinated clock, delayed release, optical or sensor substitution, shared-environment coupling, destruction and unrelated

reconstruction, data replay, or analyst leakage.

17. **Safe release** — termination of new attempts, removal or discharge of endpoint states in a stored order, return or quarantine of any unresolved custody, and preservation of all ledgers.
18. **Validity states** — qualifying, valid, degraded, out-of-envelope, invalid, quarantine, and safe release.

THEORETICAL FOUNDATION AND OPEN TRANSFER QUESTION

In the motivating POAMS framework, projected distance is a relation in an address display. If a physical boundary can establish a new completion edge between addresses A and B, the operation need not traverse each displayed intermediate address. Graphically, adjoining edge **e_AB** changes the operational cost to

d'(A,B) = min[d(A,B), w(e_AB)].

This theorem neither creates the edge nor establishes custody transfer. The open relation is

Y = M(x_A, x_B, q, χ, E, C, t),

where endpoint states **x_A,x_B**, fresh challenge **q**, isolation condition **χ**, support environment **E**, and custody class **C** produce the complete four-ledger output **Y**. M may be zero or conventional.

APPARATUS (100; FIG. 1)

The source assembly (110) and terminal assembly (120) each include a boundary coupler, local reference, state actuator, energy and momentum sensors, environmental witnesses, and tamper-evident logger. Address registries (130) bind hardware identities and permitted states. The terminal challenge generator (140) supplies a fresh value through the declared authentication channel (145). Completion controllers (150) establish challenge-bound states. Isolation witnesses (160) surround both endpoints and every permitted channel. Custody stores (170) bind unique tokens and physical invariants. The adjudicator (180) receives committed encrypted transcripts, releases blinded labels only after acquisition closes, and classifies the trial. Safe release (190) returns both endpoints to baseline.

TRIAL METHOD (FIG. 4)

The controller commits the trial schedule and analysis; verifies endpoint identities and baselines; generates a terminal challenge; sends only allowed challenge information; prepares active, wrong-address, wrong-phase, dummy, delayed, bypass, or drive-off states under blinded labels; opens a bounded completion window; records all ledgers; closes and hashes the transcripts; releases labels; tests the frozen acceptance criteria; and classifies the result as null, ordinary coupling, artifact, unpaired anomaly, paired receipt, qualified adjacency, custody transfer, or payload transfer.

FOUR-LEDGER TEST (FIG. 3)

For a paired receipt, the source and terminal events must each be present with correct challenge relation. For adjacency, ordinary channels and delay/replay countermodels must be bounded. For custody transfer, the unique physical custody record must move without duplication and the support ledger must close. Energy disappearance is not accepted as evidence; every measurable term remains in the account.

LAYERED SCOPE (FIG. 8)

- **M0:** synchronized endpoint records.
- **M1:** challenge-selective paired receipt.
- **M2:** qualified direct adjacency after isolation and countermodel gates.
- **M3:** custody transfer of an encoded retained state.
- **M4:** nonliving instrument package.
- **M5:** payload transport.

Each layer has separate qualification and does not inherit success automatically.

QUALIFICATION GATES

- **M1 freshness:** challenge unpredictability and trial binding are verified;
- **M2 selectivity:** right-address/right-phase trials differ from wrong-address, wrong-phase, and dummy trials;
- **M3 isolation:** ordinary electromagnetic, optical, acoustic, mechanical, thermal, software, timing, and shared-environment channels are bounded below the event;
- **M4 paired timing:** endpoint receipts satisfy the frozen local-clock relation without post-hoc window selection;
- **M5 support closure:** energy, momentum, angular momentum, heat, and control accounts reconcile;
- **M6 custody uniqueness:** no duplicate or substitute exists and source/terminal custody transitions reconcile;
- **M7 countermodels:** delayed release, pre-positioned duplicate, destruction/reassembly, replay, and analyst leakage fail blinded controls;
- **M8 tariff mapping:** distance, phase, mismatch, aperture, complexity, and energy dependence are measured rather than assumed;
- **M9 safe interruption:** endpoint or channel failure enters safe release or quarantine; and
- **M10 independent confirmation:** a second apparatus, endpoint pair, or laboratory reproduces the scoped result.

EMBODIMENTS

E1—resonant paired-receipt bench. Two shielded resonator or oscillator assemblies use an ordinary authenticated challenge link and a physically separate attempted-completion channel. Terminal detectors record phase, amplitude, energy, timing, and environmental witnesses. The apparatus is useful for isolation metrology even at null.

E2—address challenge array. One source is tested against plural candidate terminals, only one of which presents the fresh correct challenge. Trials randomize correct address, near-match address, wrong phase, dummy load, and terminal absence.

E3—custody-token transfer. A unique physical state, one-time hardware token, trapped excitation, or other retained state is prepared at the source. Terminal acquisition is accepted only when source possession ceases according to the frozen rule and destructive readout, copying, and data replay are excluded.

E4—paired aperture. Arrays at both endpoints create a bounded aperture with admission and release interlocks. Capacity is measured as a tariff surface; uncontrolled aperture growth is a safe-release fault.

E5—prophetic instrument-package transfer. A nonliving instrument package carries clocks, inertial sensors, memory challenges, environmental records, and tamper witnesses. The package must return with its internally generated records and no source duplicate before any payload claim.

PROPHETIC EXAMPLES

P1—two-laboratory receipt test. Laboratories share only the declared challenge link and synchronized protocol. Independent network monitors, RF spectrum analyzers, optical detectors, accelerometers, microphones, thermal sensors, and power meters bound ordinary coupling. The result may be a null.

P2—distance and tariff sweep. The same frozen endpoint design is tested at increasing ordinary separation while controlling latency, network path, and environment. No distance-independent law is presumed.

P3—custody ladder. Only after M1–M10 success does the program proceed from a one-time physical state to a passive instrument, then an autonomous instrument package, and later a payload under separate safety review.

POAMS-NATIVE DIRECT-OPERATION EMBODIMENT

In the prophetic native embodiment, endpoint readiness and a fresh challenge select a lawful completion between two already addressed boundaries. A retained organization is not transmitted through every displayed intermediate address; custody settles from source to terminal under a single co-signed operation. This mechanism is a research claim, not a represented result.

CLAIMS

1. A paired-endpoint apparatus comprising: source and terminal boundary assemblies each having an authenticated address record, a local timing reference, a state actuator or detector, energy and momentum sensing, environmental witnesses, and a tamper-evident logger; a terminal challenge generator configured to generate a fresh challenge after commitment of a trial schedule; a declared authentication channel carrying challenge information; a separately instrumented attempted-completion channel; controllers configuring challenge-bound endpoint states; isolation witnesses monitoring ordinary physical and data paths; source, terminal, support, and custody ledger stores; and an adjudicator configured to classify synchronized endpoint records under predetermined acceptance gates and to withhold a direct-adjacency or custody-transfer classification unless the corresponding ledgers reconcile.
2. The apparatus of claim 1, wherein the endpoint assemblies comprise resonant, optical, radio-frequency, acoustic, superconducting, mechanical, magnetic, electronic, or hybrid phase-bearing couplers.
3. The apparatus of claim 1, wherein the declared authentication channel is bandwidth- and energy-bounded below a claimed terminal event and is continuously monitored.
4. The apparatus of claim 1, wherein the adjudicator compares correct-address and correct-phase trials with wrong-address, near-match, wrong-phase, dummy-load, delayed-release, bypass, and drive-off trials.
5. The apparatus of claim 1, wherein trial transcripts are committed or cryptographically hashed before blinded labels are released.
6. The apparatus of claim 1, wherein the custody ledger binds a unique physical token and rejects simultaneous source and terminal possession.
7. The apparatus of claim 1, wherein a calibration memory stores a tariff surface as a function of distance, phase, mismatch, aperture, payload complexity, preparation time, and support energy.
8. The apparatus of claim 1, further comprising a safe-release controller that removes endpoint states in a stored order and quarantines unresolved custody.
9. A method of testing an address-selective paired completion, comprising committing a randomized trial schedule and analysis; authenticating source and terminal address records; generating a fresh terminal challenge; conveying challenge information through a declared bounded channel; preparing endpoint physical states bound to the challenge; opening an attempted-completion window; synchronously recording source, terminal, support, custody, isolation, and environmental channels; closing and committing trial transcripts; releasing blinded labels; comparing active trials with wrong-address, wrong-phase, dummy, delayed, bypass, and drive-off controls; and classifying the result without treating an unpaired endpoint event as direct adjacency or payload transport.
10. The method of claim 9, further comprising measuring and bounding electromagnetic, optical, acoustic, mechanical, thermal, software, timing, and shared-environment paths.

11. The method of claim 9, further comprising requiring an independently generated terminal event whose challenge relation was frozen before the attempted-completion window.
12. The method of claim 9, further comprising reconciling energy, momentum, angular momentum, heat, and control traffic in the support ledger.
13. The method of claim 9, wherein a paired receipt is classified as direct adjacency only after hidden-channel, duplicate, delayed-release, substitution, destructive-reconstruction, replay, common-clock, and analyst-leakage countermodels are rejected within stated uncertainty.
14. The method of claim 9, wherein a custody transfer is classified only after terminal custody is established, source custody is extinguished or handed off under a frozen rule, no simultaneous duplicate exists, and declared retained invariants are preserved.
15. The method of claim 9, further comprising repeating trials at plural separations and estimating a tariff surface without presuming distance independence.
16. A paired-aperture transport system comprising plural source boundary cells, plural terminal boundary cells, authenticated address and challenge processors, an admission controller, custody sensors, a calibration memory storing a capacity and tariff map, and a safe-release interlock, wherein the admission controller permits a transfer attempt only inside a validated aperture, complexity, energy, timing, and mismatch envelope.
17. The system of claim 16, wherein the controller progressively admits an encoded retained state, passive instrument, or nonliving instrument package according to separate qualification layers.
18. The system of claim 16, wherein uncontrolled increase in admitted aperture, occupancy, energy, or unpaired custody causes closure of the aperture and quarantine.
19. A qualification apparatus comprising at least three candidate terminal assemblies, a source assembly, a challenge generator selecting one terminal per trial, independent channel monitors, a randomized-state controller, and an adjudicator configured to determine address selectivity and write a scoped empirical adjacency map or null certificate.
20. The apparatus of claim 19, wherein at least one nonselected terminal presents a near-match challenge and at least one presents a correct challenge outside the selected timing or phase window.
21. A prophetic direct-adjacency transport apparatus comprising paired endpoint boundaries, an address selector, a co-signing readiness protocol, a closure-preserving transfer controller, a custody ledger, a support ledger, and a safe-release subsystem, wherein achieved payload class, aperture, duration, distance relation, and tariff are outputs of operation and not fixed limitations.
22. A prophetic method of operating the apparatus of claim 21, comprising selecting endpoint addresses; obtaining co-signed readiness; establishing a bounded aperture; transferring custody of a retained nonliving package; recording internally generated package data, endpoint ledgers, and support accounts; excluding duplication, substitution, destruction/reassembly, and ordinary-route transport; releasing or returning the package; and reporting a qualified transfer, ordinary mechanism, artifact, or null.

ABSTRACT

A paired-endpoint direct-adjacency apparatus uses authenticated source and terminal boundary assemblies, a fresh terminal challenge, challenge-bound physical states, a bounded authentication channel, a separately instrumented attempted-completion channel, isolation witnesses, and source, terminal, support, and custody ledgers. Randomized correct-address, near-match, wrong-phase, dummy, delayed, bypass, and drive-off trials are committed before blinded labels are released. An adjudicator distinguishes local anomalies, ordinary coupling, paired receipts, qualified adjacency, custody transfer, and payload transport; no higher classification follows without its own gates. Tariff, capacity, phase, mismatch, distance, energy, and safe-release envelopes are measured and stored. Resonant receipt tests, address arrays, custody-token systems, and paired apertures are disclosed. A prophetic embodiment transfers a retained nonliving package through a selected direct adjacency; no nonlocal completion or payload transport is represented as achieved.

Source and terminal endpoint assemblies, challenge channel, attempted completion channel, witnesses, and adjudicator

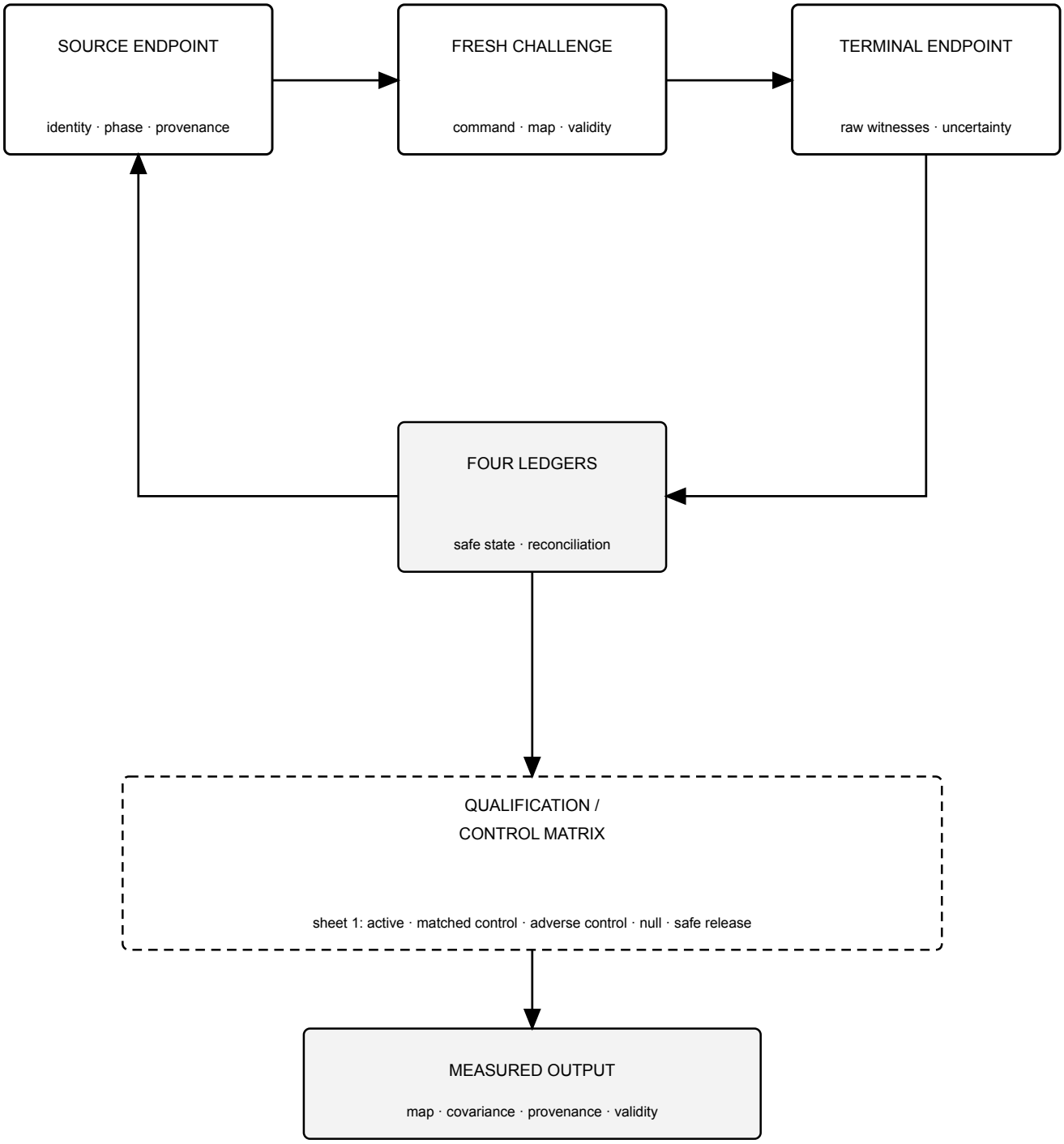


FIG. 1

Source and terminal endpoint assemblies, challenge channel, attempted completion channel, witnesses, and adjudicator.

Challenge-bound address record and trial transcript

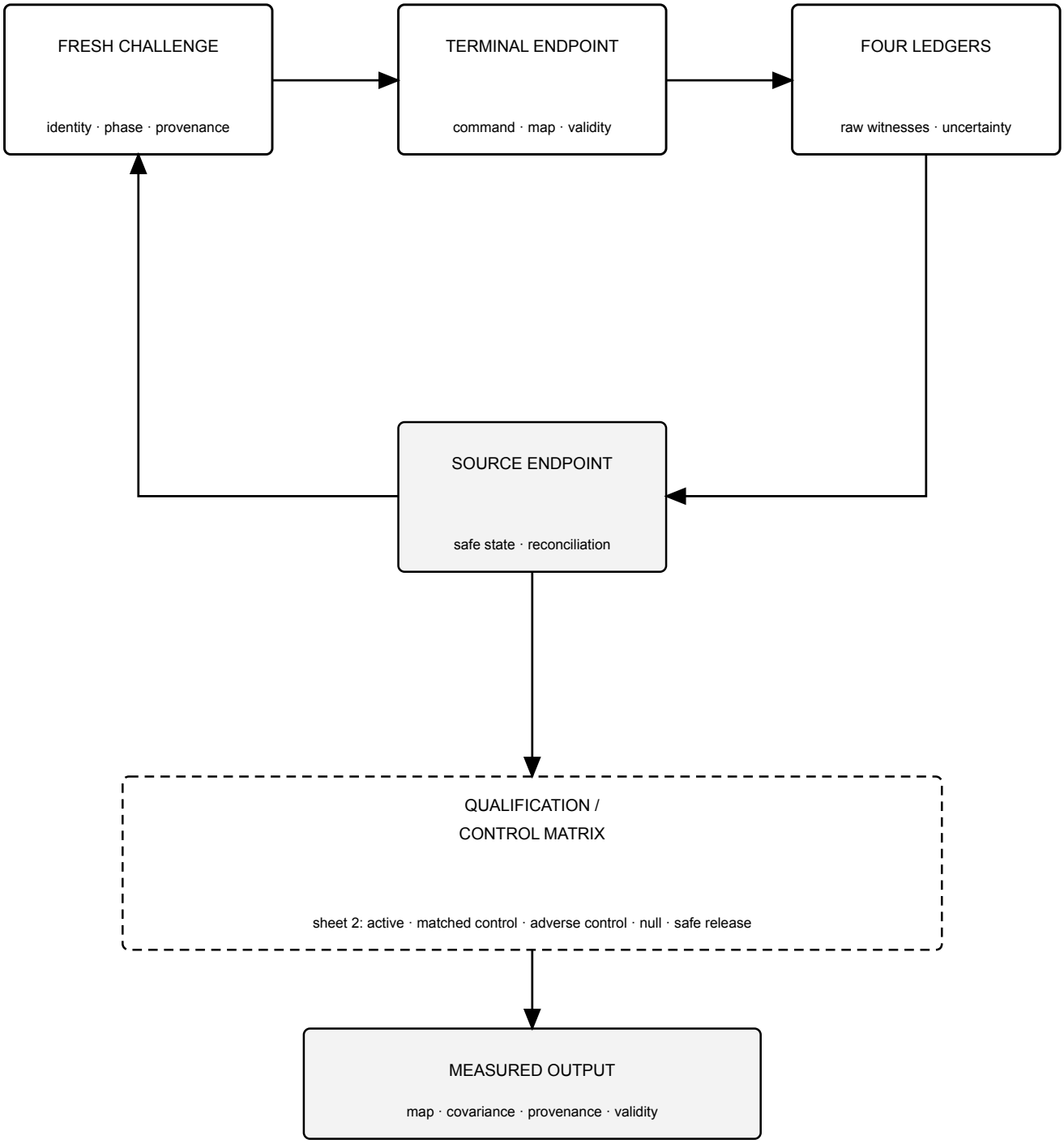


FIG. 2
Challenge-bound address record and trial transcript.

Four-ledger source, terminal, support, and custody reconciliation

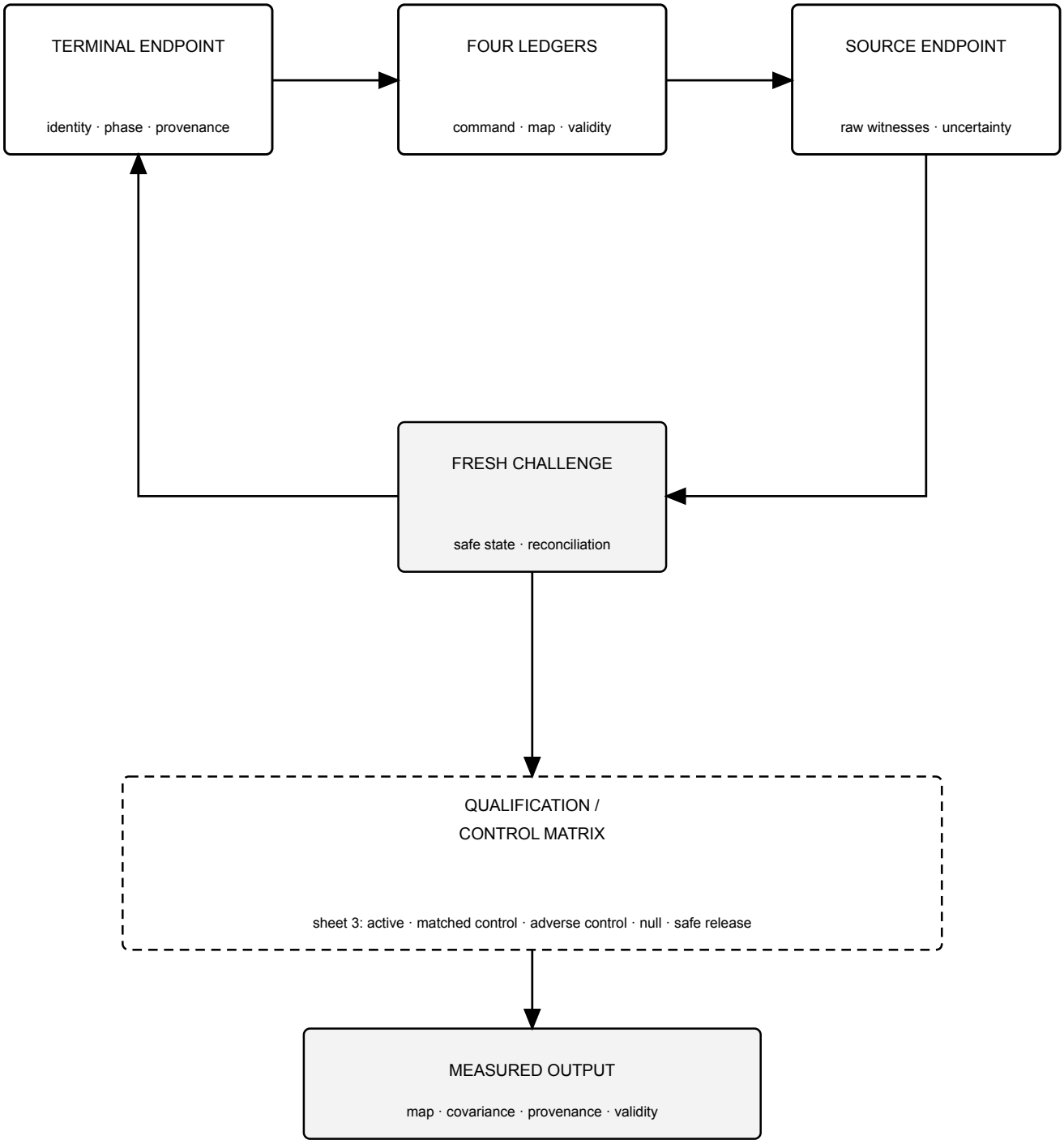


FIG. 3
Four-ledger source, terminal, support, and custody reconciliation.

Randomized trial state machine and blinded label release

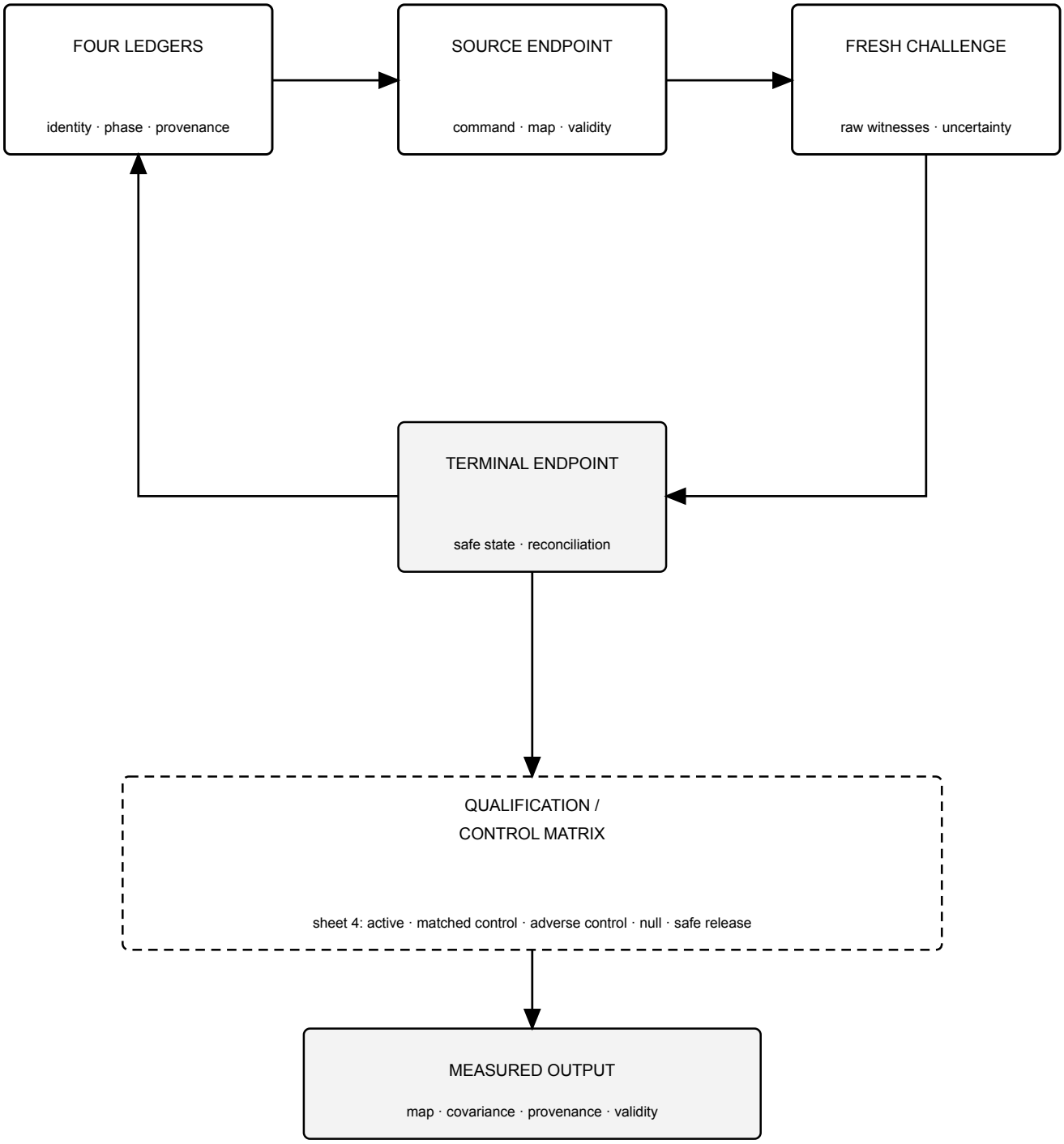


FIG. 4
Randomized trial state machine and blinded label release.

Ordinary-channel isolation enclosure and independent witness instruments

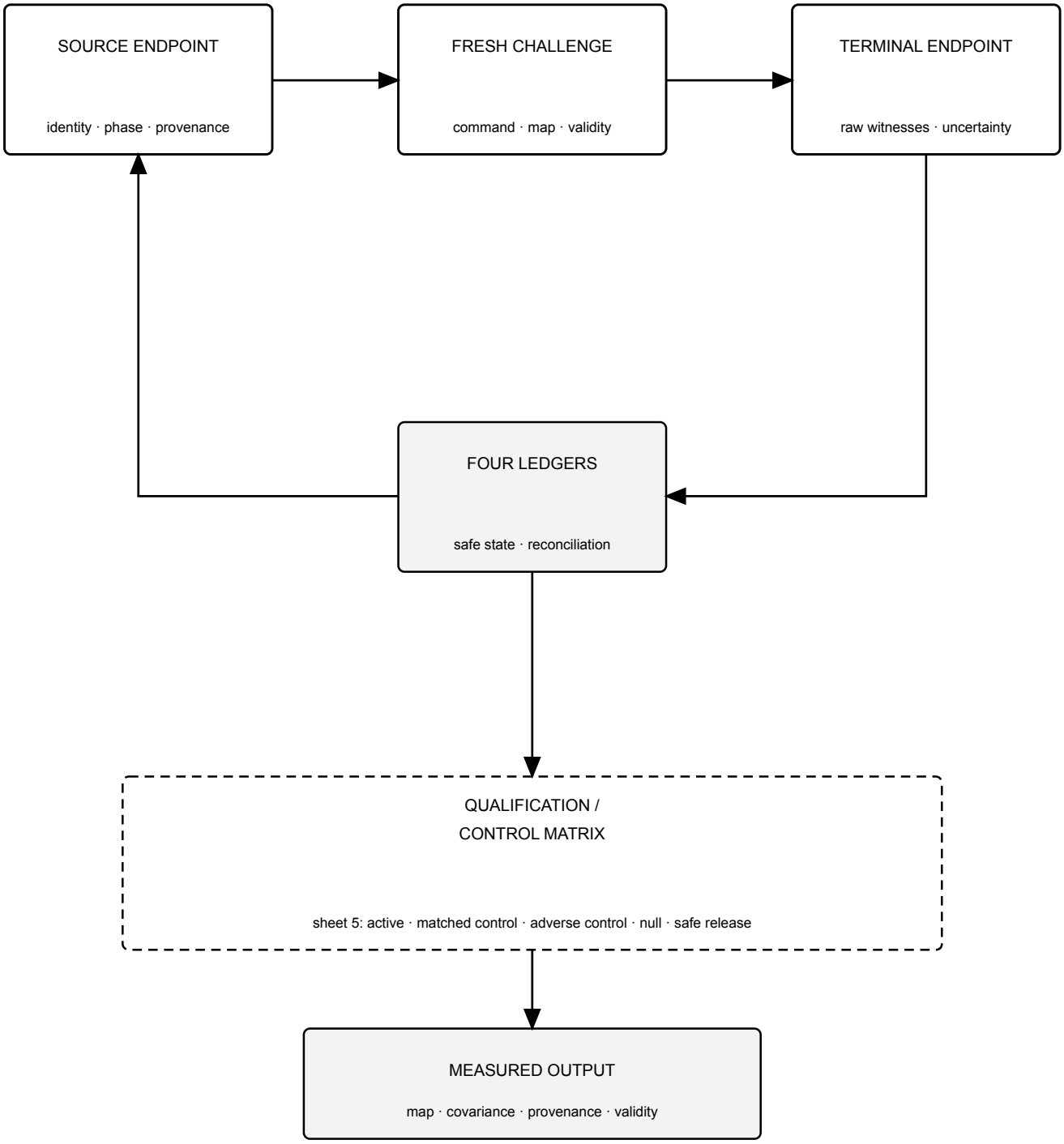


FIG. 5
Ordinary-channel isolation enclosure and independent witness instruments.

Countermodel matrix for hidden channel, duplicate, delayed release, substitution, destruction/reassembly, and clock artifact

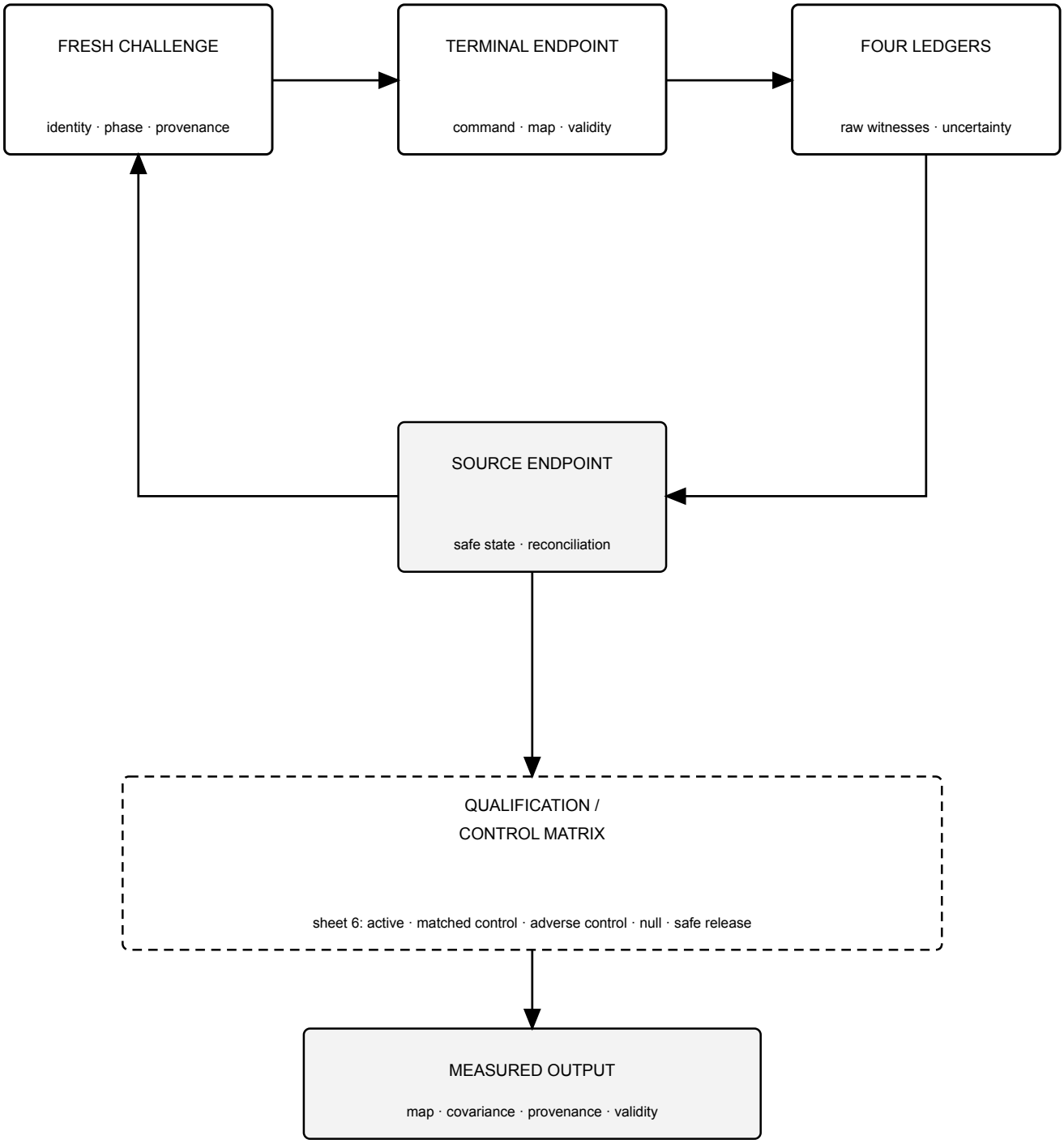


FIG. 6

Countermodel matrix for hidden channel, duplicate, delayed release, substitution, destruction/reassembly, and clock artifact.

Empirical adjacency map, tariff surface, validity envelope, and safe release

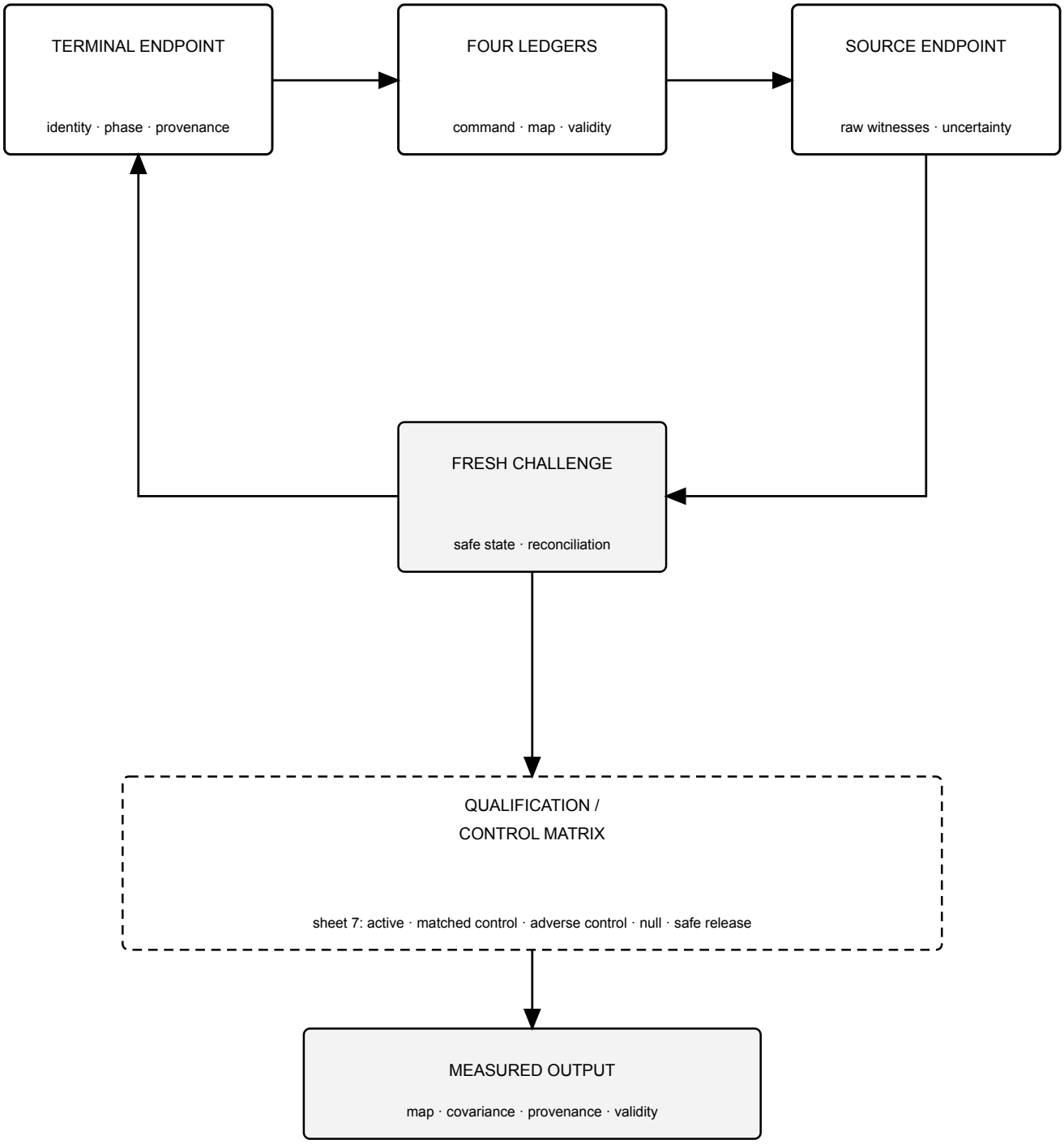


FIG. 7
Empirical adjacency map, tariff surface, validity envelope, and safe release.

Progression from receipt to adjacency, custody, instrument package, and payload

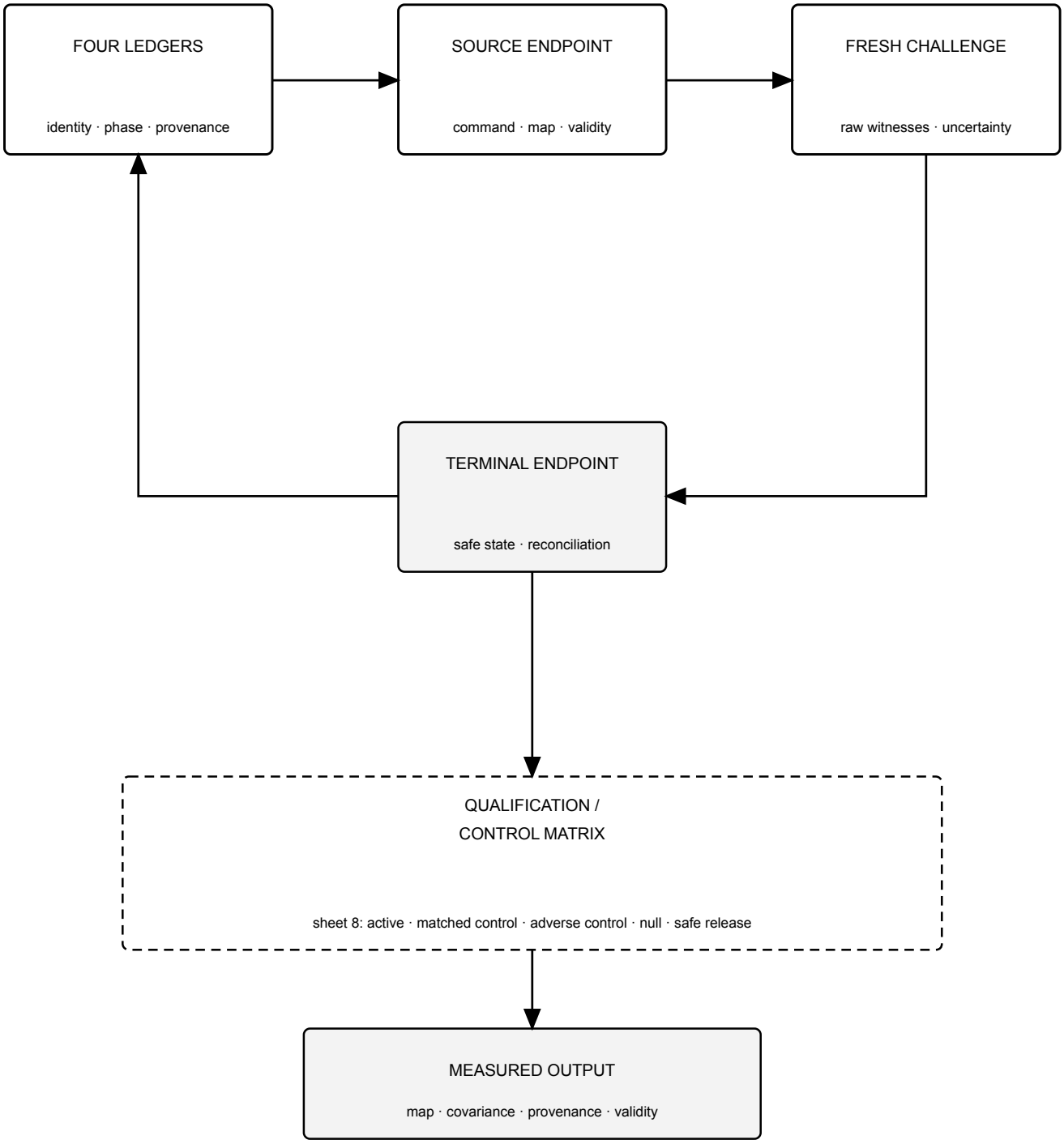


FIG. 8
Progression from receipt to adjacency, custody, instrument package, and payload.

Paired aperture modules and capacity-limited admission controller

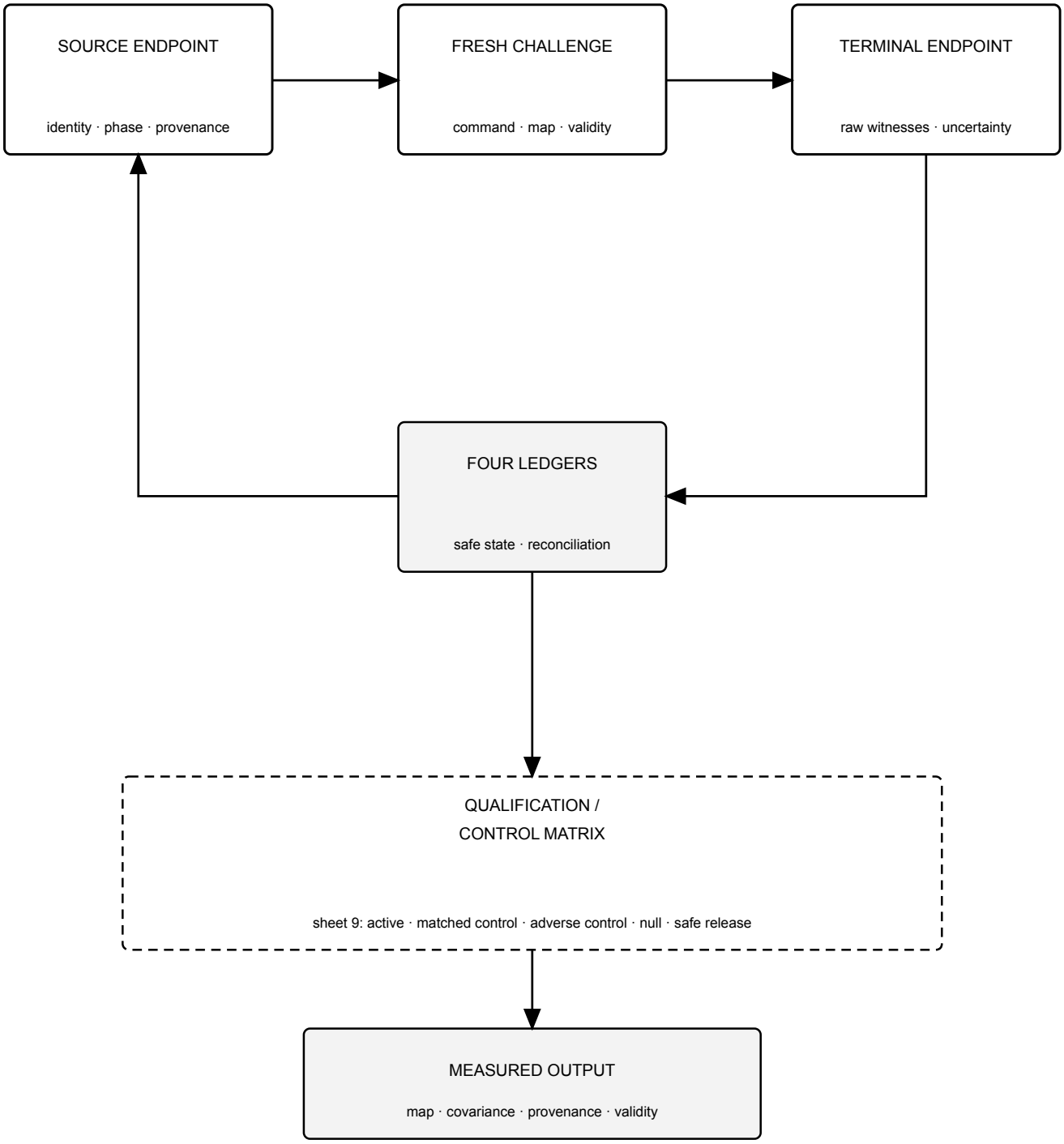


FIG. 9
Paired aperture modules and capacity-limited admission controller.

Qualification-to-deployment, provenance, validity states, and independent confirmation

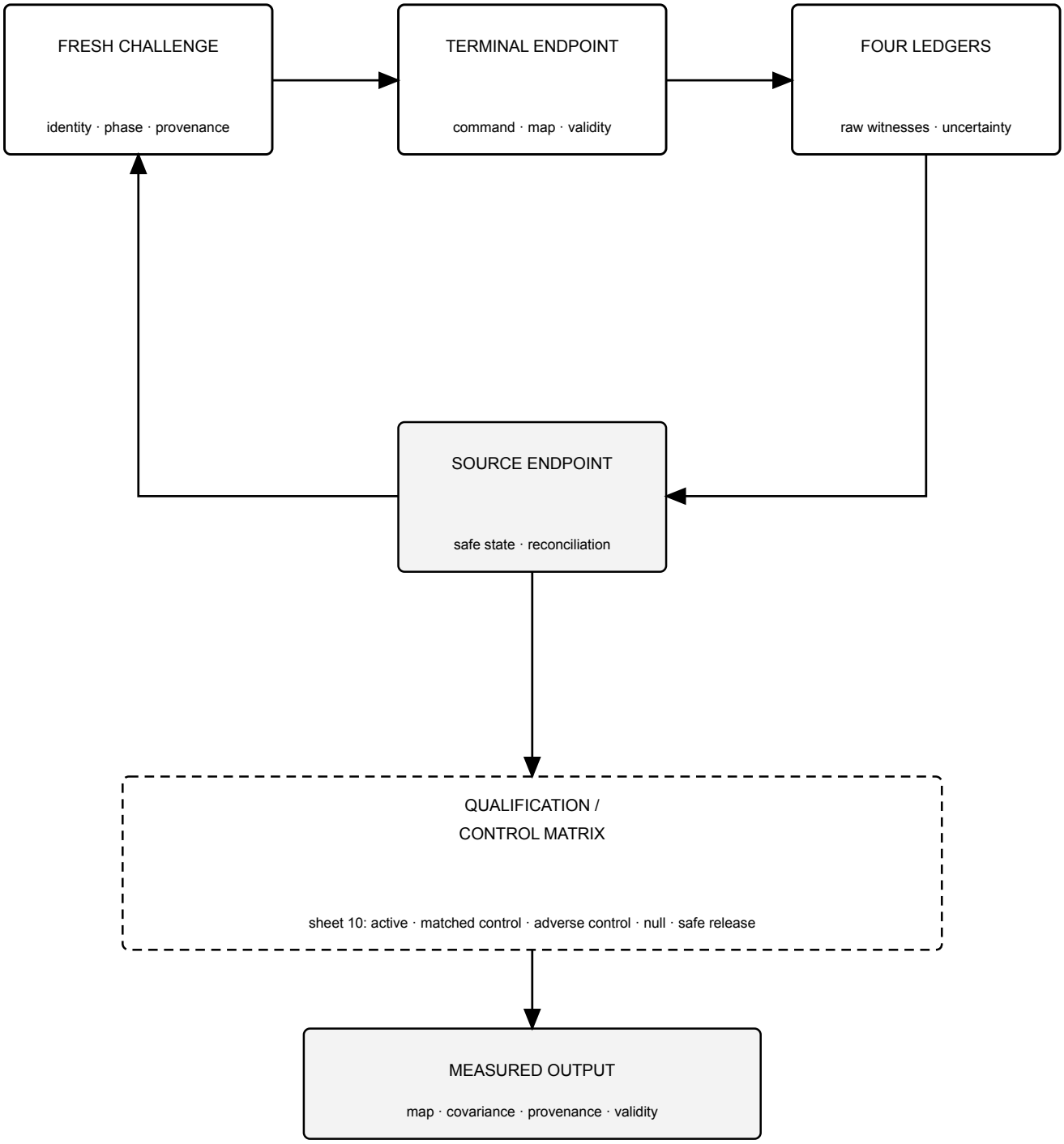


FIG. 10
Qualification-to-deployment, provenance, validity states, and independent confirmation.